

Online Counter

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Sequence - 6-1-7-2-5-3-0 ...

State Number - 7

of FlipFlop = $\log_2 7 \approx 3$

Transition table:

Present state	Next state
Y_2, Y_1, Y_0	(Y_2, Y_1, Y_0)
1 1 0	0 0 1
0 0 1	1 1 1
1 1 1	0 1 0
0 1 0	1 0 1
1 0 1	0 1 1
0 1 1	0 0 0
0 0 0	1 1 0

Excitation requirement for JK Flip Flop:

$Q_n \rightarrow Q_{n+1}$	J K
0 $\rightarrow$ 0	0 X
0 $\rightarrow$ 1	1 X
1 $\rightarrow$ 0	X 1
1 $\rightarrow$ 1	X 0

Excitation table:

Y_2, Y_1, Y_0	J_2, K_2	J_1, K_1	J_0, K_0
1 1 0	X 1	X 1	1 X
0 0 1	1 X	1 X	X 0
1 1 1	X 1	X 0	X 1
0 1 0	1 X	X 1	1 X
1 0 1	X 1	1 X	X 0
0 1 1	0 X	X 1	X 1
0 0 0	1 X	1 X	0 X

Required Equation:

$y_2 \backslash y_1 y_0$	00	01	11	10
0	1	1	0	1
1	X	X	X	X

$$J_2 = \bar{y}_1 + \bar{y}_0$$

$y_2 \backslash y_1 y_0$	00	01	11	10
0	X	X	X	X
1	X	1	1	1

$$K_2 = 1$$

$y_2 \backslash y_1 y_0$	00	01	11	10
0	1	1	X	X
1	X	1	X	X

$$J_1 = 1$$

$y_2 \backslash y_1 y_0$	00	01	11	10
0	X	X	1	1
1	X	X	0	1

$$K_1 = \bar{y}_0 + \bar{y}_2$$

$y_2 \backslash y_1 y_0$	00	01	11	10
0	0	X	X	1
1	X	X	X	1

$$J_0 = y_1$$

$y_2 \backslash y_1 y_0$	00	01	11	10
0	X	0	1	X
1	X	0	1	X

$$K_0 = \bar{y}_1$$

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